

Instructor Notes

STP Simulation Live Demo & Visualization

1. The Opening Hook (Set the Stage)

"Class, look at the topology. We have redundancy, which we want for uptime, but we have a crisis. The ticket says VLAN 10 is down. Before we touch the CLI, let's look at why this is happening visually."

2. Zoom Markup Activity: Visualizing the Storm

- **Tool:** Select the **Draw -> Line** or **Arrow** tool in Zoom.
- **Action:** Draw a bright red circle or a continuous loop between the two Gigabit lines (G0/1 and G0/2) connecting SW1 and SW2.
- **Talking Point:** "A broadcast packet enters SW1, goes to SW2, gets sent back to SW1, and it never stops. This is the 'Storm.' The switches are so busy processing these infinite loops that they can't handle actual user data from PC1."

3. The "MAC Address Trap" Markup

- **Tool:** Use the **Stamp -> X** or **Spotlight** tool.
- **Action:** Point to the MAC addresses in the CLI output of SW2.
- **Talking Point:** "Look at the MAC addresses. SW2 starts with 000a, and SW1 starts with 000c. Because we left the Priority at the default 32768, the 'Seniority' went to the lowest MAC. This is a classic CompTIA trap—the Access switch becomes the Root Bridge by accident, and our Core is now subordinate."

4. Demonstrating the Strategic Fix

- **Action:** Switch the console target to **SW1**. Type the command slowly: spanning-tree vlan 10 root primary.
- **Talking Point:** "I'm not just fixing a bug; I'm imposing a deterministic design. If we had other networks connected to the far side of this Core switch, and we let SW2 stay as the Root, every packet would have to travel 'down' to this access switch just to come back 'up' to the Core. We are making SW1 the Root to ensure the most efficient path for the entire network."

5. Final Markup: The "Logical Break"

- **Tool:** Use the **Highlighter** tool (Amber/Orange color).
- **Action:** Highlight the **G0/2** interface on the topology image.
- **Talking Point:** "Now, look at the CLI on SW2. G0/2 is now **Altn BLK**. In our diagram, visualize this link as being 'cut' logically. The cable is still there physically, but STP is refusing to send data over it to keep the network stable. We have redundancy without the loop."

6. Engagement & Check for Understanding

- "If I unplug G0/1 right now, what happens to the status of G0/2, and how long will the users be offline?"
- "Why did we apply the fix to the Core switch instead of just shutting down the port on the Access switch?"